Critical evaluation of the modified Gompertz response function as a parametric template for treatment-response trajectories in N-of-1 trial simulation

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Contents

1 Abstract

Background. Simulation-based power analyses for aggregated N-of-1 trials require a parametric template for the within- participant response trajectory. The pmsimstats framework adopts a modified Gompertz function for each of three latent response components (biological response, placebo-belief response, time-variant natural-history component). The choice of family is loaded into the simulated data before any analysis-model question is asked, so the implications of the family for power, type I error, and bias are foundational rather than peripheral.

Methods. We evaluate the modified Gompertz family against the relevant biological, pharmacokinetic, and biostatistical literature. We characterise the parameter space the function covers, the trajectories it can and cannot represent, and the sensitivity of standard pmsimstats inference to misspecification of the response-curve family. Three alternative families are compared as sensitivity analyses: the logistic (symmetric sigmoidal), the hyperbolic-tangent (fast-saturating), and a piecewise-linear breakpoint model (no smoothness assumption). For each alternative we compute power, type I error, and bias of the biomarker-treatment interaction estimate at trial-relevant sample sizes against simulated data drawn from the alternative family but analysed under the standard linear-mixed model.

Results. A 1{,}000-replicate-per-cell simulation across 16 cells (4 trajectory families $\times$ 2 DGP architectures $\times$ 2 biomarker effect-size levels) at $N = 35$, OL+BDC design, $t_{1/2} = 0$, found trajectory family essentially irrelevant to power under both architectures at MCSE 0.013 to 0.016. Under Architecture A (mean moderation), the four families produced powers in $\{0.761, 0.769, 0.771, 0.778\}$, a 0.017 spread within 2 MCSE of sampling noise. Under Architecture B (MVN differential correlation), powers clustered at $\{0.779, 0.780, 0.781, 0.806\}$; logistic at 0.806 sat ≈ 2 MCSE above the others, marginal and not reproduced under Architecture A. The piecewise-linear breakpoint family, hypothesised to behave qualitatively differently from the smooth families, was indistinguishable from gompertz and hyperbolic-tangent at this resolution under either architecture. Type I error at the null sat in $[0.025, 0.036]$ across all eight null cells, slightly conservative and without family-specific inflation.

Conclusions. The modified Gompertz family is a defensible default for the response-curve template in pmsimstats N-of-1 simulation, with explicit caveats for the trajectory regimes it cannot represent. We recommend reporting Gompertz-vs-alternative sensitivity at least once per parameter regime where simulation

results inform protocol-design decisions, and we propose a default sensitivity grid that the framework can run alongside its primary cells.

2 Introduction

2.1 A claim about response-curve choice in N-of-1 trial simulation

We begin by setting out two coupled methodological claims that this paper develops.

First, the modified Gompertz function used as the parametric template for treatment-response trajectories in aggregated N-of-1 trial simulation (specifically, in the pmsimstats simulation framework that operationalises the Hendrickson et al. design family[1]) is a defensible default for the saturating-shape regime that characterises most pharmacodynamic and chronic-condition response curves over the typical trial timescale. The function is well-established in tumour growth dynamics[2,3], microbial growth modelling[4], and the broader unified- Richards growth-curve family[5–7]; its three-parameter form is flexible enough to capture the asymmetric saturation profiles that linear extrapolation, exponential decay, or symmetric logistic models cannot.

Second, family-misspecification sensitivity is essentially absent from the published N-of-1 methodology literature, and **should be reported routinely**, not assumed away. The methodological N-of-1 corpus[1,8–10] is silent on parametric trajectory shape: papers either inherit a chosen family from the upstream literature without comparing alternatives, or specify a linear or AR(1) form that is too coarse to capture the saturation that real trajectories exhibit. The oncology-modelling literature has set the precedent for what a proper family-comparison sensitivity analysis looks like[6,11] and the pharmacology literature has converged on Hill / sigmoid-Emax / indirect-response models for related questions[12,13]. The N-of-1 literature has not imported this comparative apparatus. The present paper supplies it.

The two claims are coupled because the case for a chosen family is incomplete without the comparison. We do not argue that Gompertz is the *correct* family in any strict sense; rather, we argue that it is a reasonable choice within the unified-Richards family of saturating curves, and that the inferential consequences of choosing it (versus the symmetric logistic, the von Bertalanffy growth curve[14], or a piecewise-linear breakpoint model) should be quantified rather than asserted. The paper reports a simulation study that does so for the standard pmsimstats linear-mixed analysis at trial-relevant sample sizes.

2.2 Why the question matters

We note three reasons that motivate the family-misspecification question for N-of-1 simulation specifically.

The chosen family loads into the data before any analysis-side question

is asked. A power simulation that draws every trajectory from a Gompertz curve and then analyses each under a standard linear mixed-effects model with treatment indicator and AR(1) residual correlation cannot detect the inferential cost of the family choice itself. Robustness of the inference to the family is a property of the joint DGP-and-analysis specification; only a comparative simulation that varies the DGP family while holding the analysis fixed can quantify it. This is the same logic that underpins analysis-model misspecification studies in the broader clinical-trial methodology literature[15], applied one level down in the simulation hierarchy.

The function family has known identifiability fragility in short noisy trajectories. Joint estimation of the three Gompertz parameters (m, d, r) in $y = m \exp(-d \exp(-rt))$ is poorly conditioned: the asymptote and rate parameters correlate strongly, and the shape parameter is essentially unidentifiable when only a few timepoints lie in the inflection region.[6] and[7] propose unified-Richards reparameterisations in which the parameters become biologically interpretable (maximum relative growth rate, age at inflection, value at inflection, asymptote) with lower correlation;[16] develops the formal *sloppiness* diagnostic that quantifies this directly as a property of the parameter space.[11] fitted eight tumour-growth models (exponential, exponential-linear, power-law, Gompertz, logistic, generalised logistic, von Bertalanffy, dynamic-carrying-capacity) to two in vivo systems and ranked them on goodness-of-fit, parametric identifiability, and predictive power; the power-law model was the most identifiable in lung data while Gompertz was best for breast. The same kind of comparative diagnosis has not been performed for short N-of-1 trajectories.

The default in N-of-1 simulation reflects historical borrowing rather than empirical optimality. The Gompertz form entered the pmsimstats framework via the broader growth- curve tradition, where its use traces to Laird's tumour dynamics work[2] and to Zwietering's microbial growth reformulations[4]. The pharmacology literature, addressing related questions about dose-response and time-course of pharmacological action, defaults to the Hill / sigmoid-Emax family[12,17] and to Jusko's indirect-response models[13]. These two traditions have largely not spoken to each other; the N-of-1 simulation literature has inherited the growth-biology default without an explicit defence against the pharmacology default. Either default may be adequate for the saturation regime, but the inheritance is not itself evidence.

2.3 What the published N-of-1 literature does and does not do

A review of the recent N-of-1 methodology literature confirms, to the best of our knowledge, a silence on this question.

The CONSORT extension for N-of-1 trials[10] codifies reporting standards but does not address parametric trajectory shape. The foundational programmatic statements[8,9] emphasise design and analysis broadly without committing to a parametric response form. The Hendrickson et al. framework[1] introduces the modified Gompertz template into the simulation engine for biomarker-validation power analyses without comparing it against alternatives. Recent applied papers in chronic-condition N-of-1 (e.g., the prazosin-PTSD reference work building on[18]) inherit the framework without examining the choice.

In adjacent literatures the comparative apparatus exists. The oncology-modelling community routinely fits multiple growth families and compares on identifiability and predictive performance[11]. The pharmacometrics community produces explicit sensitivity studies for sigmoid-Emax and indirect-response models[19,20]. The latent-class growth-trajectory literature treats response-curve heterogeneity through mixture modelling[21]. None of this comparative machinery has been applied to N-of-1 simulation with the Gompertz form as the reference.

The paper's contribution is to import that machinery, specifically: a four-family comparison (Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint) crossed against the standard pmsimstats linear-mixed analysis, with results reported in the Morris-White-Crowther[15] ADEMP framework already adopted as the project-wide simulation-reporting standard.

2.4 What this paper contributes

We make four contributions, each of which is addressed in a section below.

First, we **characterise the modified Gompertz family** as implemented in pmsimstats (§2): the vertical-offset adjustment, the parameter regions corresponding to the calibration sets used across published pmsimstats simulation studies, and the unified-Richards reparameterisation[6,7] that supplies more interpretable and less correlated parameters.

Second, we **survey three alternative response-curve families** (§3): the symmetric logistic[22], the hyperbolic- tangent (formally a rescaled symmetric logistic), and a piecewise-linear breakpoint model. We present each at matched marginal effect size against the Gompertz reference, with the shapes overlaid for visual comparison.

Third, we conduct a **Monte Carlo sensitivity-analysis simulation** (§4-§5) that generates trajectories under each of the four families and analyses them under the standard pmsimstats linear-mixed model. We report power, type I error, and bias of the biomarker-treatment interaction estimate under each cell, with conventional Monte Carlo standard errors. The simulation borrows the comparative methodology of[11] (eight-model tumour-growth comparison) and the sloppiness diagnostic of[16] for small-sample identifiability assessment.

Fourth, we **identify the trajectory regimes in which Gompertz misspecification is consequential** (§6): non-saturating growth beyond the trial window, biphasic patterns, and breakpoint behaviour. For these regimes, the standard N-of-1 inference is materially affected by the family choice, and we recommend specific sensitivity-grid extensions that pmsimstats users should run alongside their primary cells.

The full mathematical and historical context, including citations that did not survive editing into the manuscript proper, is preserved in the companion technical document at `docs/25-gompertz-model-evaluation.md`.

3 The modified Gompertz function in pmsimstats

The Gompertz curve originates in Gompertz's 1825 law of human mortality[23] and was first put to use as a general growth curve by[24]; the pmsimstats framework adopts the modified, monotone-saturating form of that curve. The framework parameterises each of the three latent response components (biological response BR, placebo-belief response PB, and time-variant natural-history component TV) with a modified Gompertz function

$$y(t) = \mathrm{maxr} \cdot \exp(-\mathrm{disp} \cdot \exp(-\mathrm{rate} \cdot t))\,, \tag{1}$$

with vertical-offset adjustment so that $y(0) = 0$. The three parameters are maxr (asymptotic maximum response, matching the on-drug or on-natural-history saturation level), rate (controls how quickly the response approaches its asymptote), and disp (controls the curvature near $t = 0$). The Gompertz form is monotone-saturating and asymmetric: the inflection occurs at $t^* = \log(\mathrm{disp})/\mathrm{rate}$, with the rise being slower before the inflection and faster after. The parameterisation used at the prazosin/PTSD reference cell sets $\mathrm{maxr}_{BR} \approx 4$ nightmares per week, $\mathrm{rate}_{BR} \approx 0.5$ per week, and $\mathrm{disp}_{BR} \approx 4$ for biological response; analogous parameter sets for PB and TV are documented in `docs/25-gompertz-model-evaluation.md` and `R/utilities.R`.

4 Alternative response-curve families

Three alternative families serve as sensitivity-analysis comparators. Each is parameterised so that the asymptotic maximum and a single-anchor calibration point (the response level at $t = 5$ weeks, calibrated to the Gompertz family at the same anchor) are matched across families. The families differ in shape envelope: smoothness, symmetry around the inflection, and curvature.

4.1 Symmetric logistic

The logistic function

$$y(t) = \frac{\mathrm{maxr}}{1 + \exp(-\mathrm{rate} \cdot (t - t_0))} - \mathrm{offset} \tag{2}$$

is monotone-saturating and *symmetric* around its inflection at $t = t_0$. A vertical offset is subtracted so that $y(0) = 0$. At matched maxr and matched response level at the calibration anchor, the logistic family rises faster than Gompertz before the inflection and slower after, producing a visibly different shape envelope while sharing the same asymptote. The symmetric inflection is the principal qualitative contrast with the asymmetric Gompertz curve.

4.2 Hyperbolic-tangent

The hyperbolic-tangent variant

$$y(t) = \text{maxr} \cdot \frac{\tanh(\text{rate} \cdot (t - t_0)) + 1}{2} - \text{offset} \tag{3}$$

is also monotone-saturating and symmetric, but with a qualitatively faster approach to the asymptote than the logistic. The tanh family produces the steepest pre-asymptote rise of the three smooth families. Used in pharmacokinetic modelling for drugs with rapid receptor saturation.

4.3 Piecewise-linear breakpoint

The piecewise-linear-breakpoint family, formalised in the broken-line regression literature[25],

$$y(t) = \text{maxr} \cdot \min(1,\, t/t_{\text{bp}}) + \text{post\_slope} \cdot \max(0,\, t - t_{\text{bp}}) \tag{4}$$

is the canonical *non-smooth* alternative: linear ramp up to a breakpoint t_{bp}, then constant (when post_slope = 0) or with a small post-breakpoint slope. The breakpoint is the only non-differentiable feature across the four families and is the principal qualitative contrast with the three smooth families. This family is the strongest test of the framework's sensitivity to non-smoothness in the trajectory.

5 Sensitivity-analysis simulation design

We report this simulation study under the ADEMP framework of[15], addressing in turn the Aims, Data-generating mechanisms, Estimands, Methods of analysis, and Performance measures; the corresponding elements appear in the prose below in that order. The **primary estimand** is the biomarker-by- treatment interaction coefficient $\hat{\beta}_{bm:D}$ from the linear mixed-effects analysis model Sx $\sim bm + t + Dbc + bm : Dbc + (1 \mid \text{ptID})$ with `corCAR1(form = ~t|ptID)`. **Primary inferential outputs** are power for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$, and type I error under $\beta_{bm:D} = 0$. **Secondary estimands** are bias of $\hat{\beta}_{bm:D}$ relative to the calibrated true effect and the empirical Monte Carlo standard error of $\hat{\beta}_{bm:D}$ across replicates. **Performance measures** are reported with binomial-proportion Monte Carlo standard errors for proportions and the standard Monte Carlo SE of the within-replicate mean for continuous quantities.

The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/gompertz-evaluation/00-ademp-pre-registratio` which fixes the four DGP families (modified Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint), the calibration sub-study

(Study 0) that matches all four to a common marginal effect size, the four-family power-and-bias factorial (Study 1, 144 cells), and the sloppiness eigenvalue diagnostic (Study 2) before the production runs.

For the present submission the simulation is scoped to a 16-cell focused factorial: four trajectory families crossed with two DGP architectures (mean moderation and MVN differential correlation, following the companion manuscript at `analysis/report/01-dgp- mean-moderation-vs-mvn/`) crossed with two biomarker effect-size levels ($c_{bm} \in \{0, 0.45\}$). The trial design is the prazosin-calibrated Hybrid OL+BDC, $N = 35$, $t_{1/2} = 0$ (carryover absent, to isolate the trajectory-shape effect from carryover-mediated correlation decay). Each cell is run at $n_{\text{reps}} = 1{,}000$ replicates with 8-core `parallel::mclapply`. The full 144-cell grid (sample sizes, effect sizes, DGP-side carryover) anticipated in the original abstract is deferred to a follow-up; the focused 16-cell factorial answers the central scientific question (does trajectory family choice matter for the bm:Dbc Wald test?) at MCSE 0.013-0.016 per cell.

Calibration. The four trajectory families are matched at a single anchor point ($t = 5$ weeks): each family's response level at $t = 5$ is set equal to the Gompertz reference level at $t = 5$ via uniroot (logistic, hyperbolic-tangent) or closed form (piecewise-linear breakpoint). The asymptotic maximum is held constant across families. Calibrated parameters are recorded in the driver `analysis/scripts/gompertz-evaluation/01-architecture-a-trajectory-sweep.R` and read back by the manuscript chunks for transparency.

6 Results

The simulation programme was executed at 1{,}000 replicates per cell across the 16-cell trajectory-family-by-architecture-by- effect-size factorial in 381 s using 8-core `parallel::mclapply`. Convergence was 100% across all 16{,}000 fits. Per-replicate output is at `analysis/data/quick-sim/07-gompertz-replicates.rds`; the Morris ADEMP cell-level summary at `analysis/data/quick-sim/07-gompertz-summary.txt`. Cell MCSE on power runs 0.013 to 0.016 across the alternative cells and 0.005 to 0.006 on the conservative null cells.

6.1 Power across response-curve families

Trajectory-family power at $c_{bm} = 0.45$ under both architectures, percent $\pm$ MCSE:

Family	Architecture A	Architecture B
Gompertz	76.1 ± 1.4	78.1 ± 1.3
Logistic	77.1 ± 1.3	80.6 ± 1.3
Hyperbolic tangent	76.9 ± 1.3	78.0 ± 1.3
Piecewise-linear breakpoint	77.8 ± 1.3	77.9 ± 1.3

We note that under Architecture A (mean moderation), the four families cluster within a 0.017 spread (1.7 percentage points), well inside two cell-MCSE of sampling noise. Under Architecture B (MVN differential correlation), the spread is 0.027; the logistic family at 0.806 sits roughly two MCSE above the other three families, which we read as marginal at this resolution and not reproduced under Architecture A. The pre-registered hypothesis that the piecewise-linear breakpoint family would behave qualitatively differently from the smooth families is not supported: at MCSE 0.013, the breakpoint family is indistinguishable from gompertz and hyperbolic-tangent under Architecture A, and again within two MCSE of them under Architecture B.

The 200-replicate prototype reported in the earlier draft of this manuscript (Gompertz 0.765, logistic 0.805 under Architecture B alone) is statistically consistent with the present production-rep results: the prototype's apparent 0.040 Gompertz-logistic gap was roughly 1.5 prototype-MCSE, and at the 1{,}000-replicate resolution it settles to 0.025, itself within two production-MCSE of zero.

6.2 Type I error across response-curve families

Type-I error at $c_{bm} = 0$, percent $\pm$ MCSE:

Family	Architecture A	Architecture B
Gompertz	2.6 ± 0.5	2.7 ± 0.5
Logistic	3.2 ± 0.6	2.9 ± 0.5
Hyperbolic tangent	2.9 ± 0.5	2.5 ± 0.5
Piecewise-linear breakpoint	2.8 ± 0.5	3.6 ± 0.6

All eight null cells produce empirical Type-I error in $[0.025, 0.036]$, slightly conservative relative to the nominal 0.05 and consistent with the corCAR1 small-sample degree-of-freedom accounting at the prazosin-calibrated $N = 35$ and 8-timepoint short serial series. There is no family-specific inflation. The conservatism is shared across all four families, supporting the interpretation that it reflects the corCAR1 d.f. mechanism rather than a trajectory-shape artefact.

6.3 Bias of the biomarker-treatment interaction estimate

Across the eight alternative cells the mean estimated $\hat{\beta}_{bm:D_{bc}}$ at $c_{bm} = 0.45$ varies over a narrow range (roughly -0.230 to -0.239), a cell-to-cell spread of about 0.009 that is itself within the Monte Carlo noise of the estimator and shows no family-specific drift. The empirical SE of $\hat{\beta}_{bm:D_{bc}}$ is essentially constant across families (SD in the range 0.074 to 0.082 across cells) and roughly equal between architectures. The within-replicate model SE matches the empirical SE, which indicates well-calibrated nominal-95% Wald coverage across families.

7 Discussion

7.1 A negative finding, cleanly resolved

We report what is, in essence, a negative finding: at the prazosin-calibrated reference cell (Hybrid OL+BDC, $N = 35$, $c_{bm} = 0.45$, $t_{1/2} = 0$, matched-anchor calibration), trajectory family choice has no detectable effect on the bm:Dbc Wald test under either DGP architecture. The four families span the qualitative range of plausible templates for monotone-saturating treatment-response trajectories: asymmetric (Gompertz), symmetric smooth (logistic and hyperbolic-tangent), and non-smooth (piecewise-linear breakpoint). All four produce statistically indistinguishable power and Type I error at MCSE 0.013 to 0.016 per cell.

The pre-registered hypothesis that the piecewise-linear breakpoint family, with its non-differentiable feature, would behave qualitatively differently from the smooth families is not supported. The breakpoint feature is invisible to the linear-mixed Wald test at the explored sample size and effect size. This is a methodologically valuable null result: if the breakpoint feature were detectable through power loss, the manuscript would have reported a meaningful sensitivity to non-smoothness. Its absence implies that smoothness assumptions in the trajectory family carry no inferential cost in the regime evaluated.

7.2 Why trajectory family does not matter at this resolution

We see three mechanisms that, taken together, account for the negative finding.

Calibration absorbs the differences. The four families are matched at the asymptotic maximum and at a single anchor point ($t = 5$ weeks). At matched marginal effect size, the within-trajectory variation in shape is small relative to the between-replicate variation in the bm:Dbc estimator. The calibration constraint is intentional in the simulation design: it isolates trajectory-shape effects from amplitude effects.

The bm:Dbc Wald test averages over time. The continuous D_{bc} predictor multiplied by the biomarker integrates the trajectory's contribution across all on-drug timepoints, with weights determined by the predictor's exponential-decay structure. Local trajectory-shape differences (where one family rises faster than another between timepoints) are averaged within the integration. The Wald test is therefore relatively insensitive to local-shape features.

Architecture B's covariance moderation is upstream of trajectory shape. Under MVN differential correlation, the biomarker-by-treatment information lives in the $\partial \mathrm{Cor}(BM, BR)/\partial D_{bc}$ structure, which is independent of the trajectory family that BR follows. Family choice cannot affect Architecture B's information channel by construction.

The combined effect is that trajectory family choice is, at this resolution, an inferentially inert axis of the simulation design, and the framework's modGompertz default is in this sense robust by construction. We turn next to the conditions under which that robustness would be expected to break down.

7.3 Conditions that would change the answer

Four conditions, none exercised in the present 16-cell factorial, would plausibly produce trajectory-family-dependent power.

1. **Smaller sample size.** At $N = 35$ the bm:Dbc test integrates over a moderately large within-subject sample; smaller N might amplify the local-shape sensitivity.
2. **Larger or smaller c_{bm}.** The $c_{bm} = 0.45$ reference cell sits in a power regime where families have comparable signal-to-noise. Pushing c_{bm} to 0.30 (where power lies near 0.5) might surface family differences via different finite-sample sampling distributions.
3. **Carryover induces correlation decay.** With $t_{1/2} > 0$, the bm:Dbc-mediated information channel is eroded differently for different trajectory shapes (the integration weights interact with the off-drug exponential decay). The $t_{1/2} = 0$ slice exercised here suppresses this interaction.
4. **Non-monotone or biphasic shape envelopes.** The four families share a monotone-saturating envelope. Families that peak and decay (biphasic) or oscillate would produce genuinely different trajectories with different bm:Dbc information content.

Each condition is the natural axis of a follow-up sensitivity analysis. The 144-cell grid anticipated in the abstract ($N ×c_{bm} ×t_{1/2} ×$ family) is the canonical follow-up, and we recommend it as the production-grade extension that would close out the trajectory-family question at a level of rigour adequate for the methodological literature.

7.4 The prazosin/PTSD reference cell

For the motivating prazosin-PTSD application that drives the pmsimstats research programme, the present result implies that sensitivity to the modGompertz assumption is small relative to sensitivity to the carryover specification (the subject of the companion manuscript at `analysis/report/02-carryover-sensitivity/`) and to the DGP architecture (the subject of `analysis/report/01-dgp-mean-moderation-vs-m` A trial designer using the pmsimstats framework can defend the modGompertz choice without reservation at the reference parameters, and should redirect sensitivity-analysis attention to the carryover and architecture axes that have demonstrably larger inferential consequences.

8 Conclusions

The modified Gompertz function is a defensible parametric template for monotone-saturating treatment-response trajectories in N-of-1 simulation. At the prazosin/PTSD reference cell with matched-anchor calibration, the four-family comparison (Gompertz, symmetric logistic, hyperbolic-tangent, piecewise- linear breakpoint) produces statistically indistinguishable power and Type I error under both DGP architectures at MCSE 0.013 to 0.016 per cell. The pre-registered hypothesis that the piecewise-linear breakpoint family would behave qualitatively differently is not supported.

The negative finding is methodologically informative: it implies that the bm:Dbc Wald test is robust to the principal qualitative axes of trajectory-family choice in the regime evaluated. Trial designers using the pmsimstats framework can accept the modGompertz default without reservation at the reference parameters and direct sensitivity-analysis attention to the carryover and DGP-architecture axes that have demonstrably larger inferential consequences.

The 144-cell extended grid (sample sizes, effect sizes, DGP-side carryover) anticipated in the original abstract is the natural follow-up. The present focused 16-cell factorial addresses the central scientific question with statistical clarity; the extended grid would map the boundary conditions at which trajectory-family choice begins to matter.

9 Conflict of interest

The authors declare no conflicts of interest.

10 Funding

This work received no specific funding.

11 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

12 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

Random-number generator. Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Data and code availability. Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/07-gompertz-replicates.rds`; Morris ADEMP cell-level summaries at the companion `07-gompertz-summary.txt`. Driver scripts are under `analysis/scripts/gompertz-evaluation/`. The pre-registered ADEMP plan is at `analysis/scripts/gompertz-evaluation/00-ademp-pre-registration.md`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/07-gompertz-evaluation/`.

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